

Experience

Independent Researcher & Open-Source Developer

Feb 2026 - Present

a9l.im

- Conduct independent research from hypothesis through experiments and writeup; publish and maintain open-source projects; maintain website with simulations.

Education

University of California, San Diego

Sep 2023 - Mar 2026

B.S. Mathematics - GPA 3.75 - completed early

Projects

Saklas

Apr 2026 - Present

Mechanistic interpretability workbench - Python, PyTorch

- Built a local interpretability workbench compatible with most open source LLMs and both CUDA and MPS.
- Implemented curved manifold extraction from sets of concepts.
- Implemented Jacobian-lens readouts and nonstandard roles in chat templates.
- Published a Python API, dashboard, and OpenAI/Ollama-compatible server

Geon

Feb 2026 - Present

Relativistic N-body simulation - JavaScript, WebGPU

- Built a WebGPU simulation that models N-body physics with charged rotating massive particles.
- Implemented finite-speed force propagation and Einstein-Infeld-Hoffmann dynamics.
- Implemented force-coupled scalar fields with self-gravity.

Ogdoad

Jun 2026 - Present

Library for recreational mathematics - Rust, Python

- Built a Rust library to explore Clifford algebras over the surreal numbers and Conway's combinatorial games.
- Implemented Python tooling and designed a purpose-built language to investigate open problems.

Other Projects

Feb 2026 - Present

Browser simulations and tooling for agents - JavaScript, WebGPU

- **Coding agent infrastructure.** Gaslamp, a cross-agent CLI reviewer for Claude Code and Codex, and Dagwood, a version-controlled DAG-based task tracker with an MCP.
- **Interactive simulations.** Built seven more browser projects across physics, finance, biology, political science, and comparative religion.

Research

Introspection via Kaomoji

Apr 2026 - Jun 2026

Emotional concept representation study - Python

- Investigated correspondence between kaomoji emitted by LLMs to prompt emotional categories.
- Found that the kaomoji could predict the emotional category with 80.6% accuracy on Gemma-4-31b-it.
- Aligned hidden states across emotional category to analyze common geometry.
- Built data collection tooling and published the resulting corpus.

Elapsed-Time Encoding in LLMs

Jun 2026 - Present

Ongoing time encoding study - Python, PyTorch

- Investigated the internal representation of elapsed time within an LLM over a conversation.
- Showed elapsed time is encoded and linearly proportional to context length using probes.

Neuralese as a Conlang

Jul 2026 - Present

Ongoing abstract-CoT analysis study - Python, PyTorch

- Implemented Abstract-CoT to investigate if abstract tokens could be analyzed as a compositional language.
- A preliminary run on a small model did not successfully learn to use abstract tokens to reason.

Block Sparse Crosscoders

Jul 2026 - Present

Ongoing cross-layer representation geometry study - Python, PyTorch

- Synthesized recent work on block sparse autoencoders and sparse crosscoders to extract cross-layer multi-dimensional features.

Technical Skills

Research/ML: Python, PyTorch, NumPy/SciPy, CUDA, MPS.**Engineering:** Java, Rust, JavaScript, WebGPU, Git.